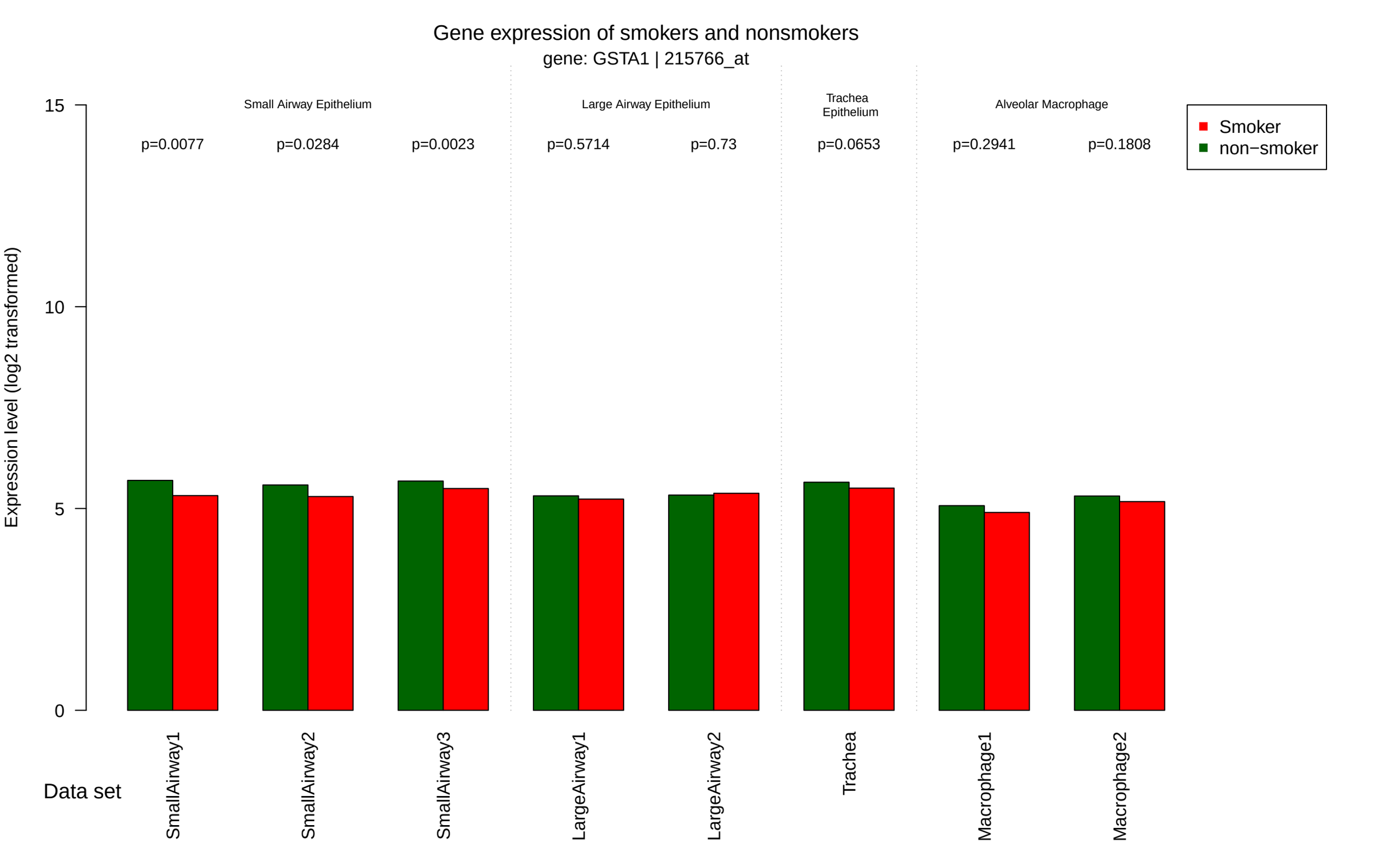
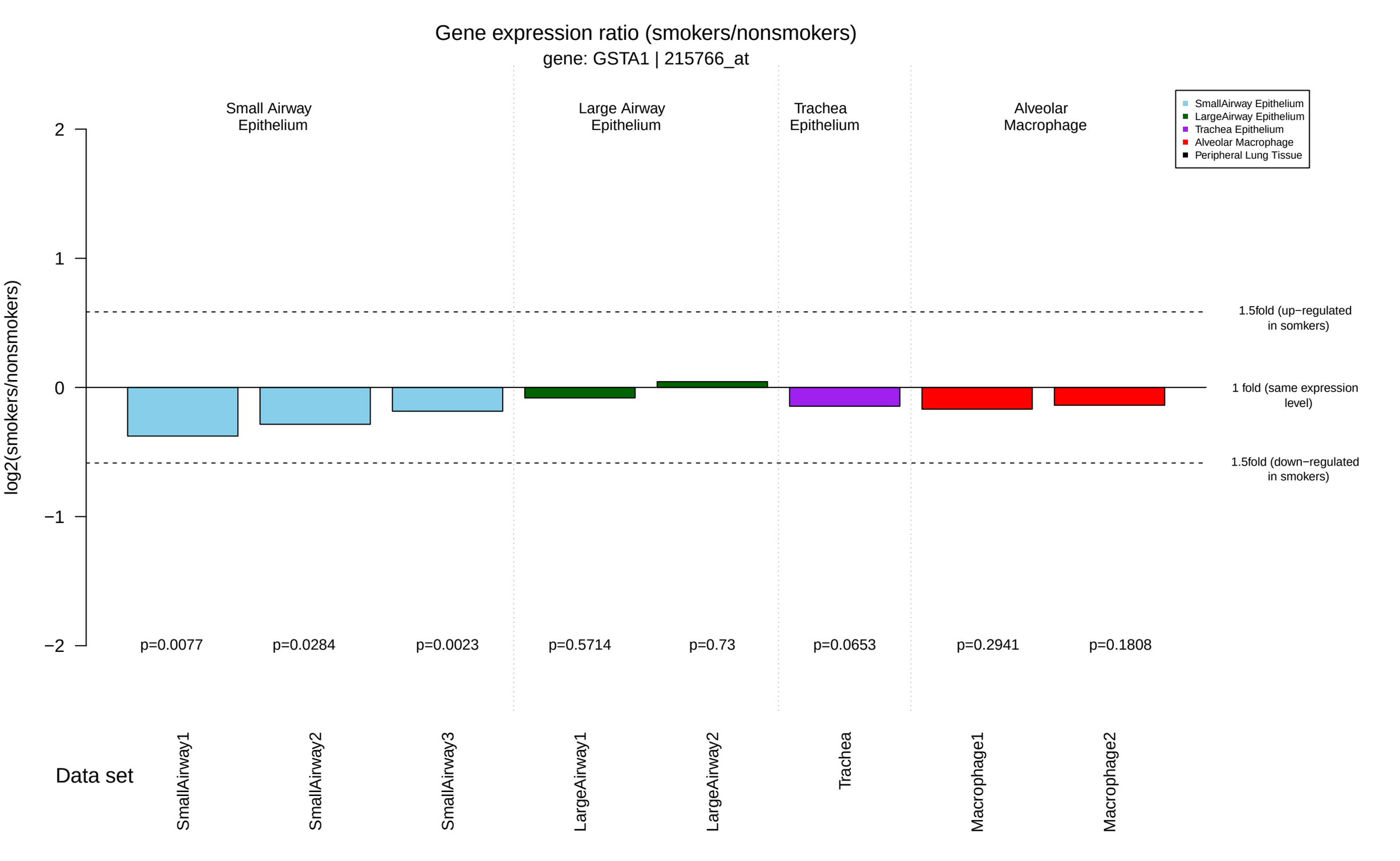
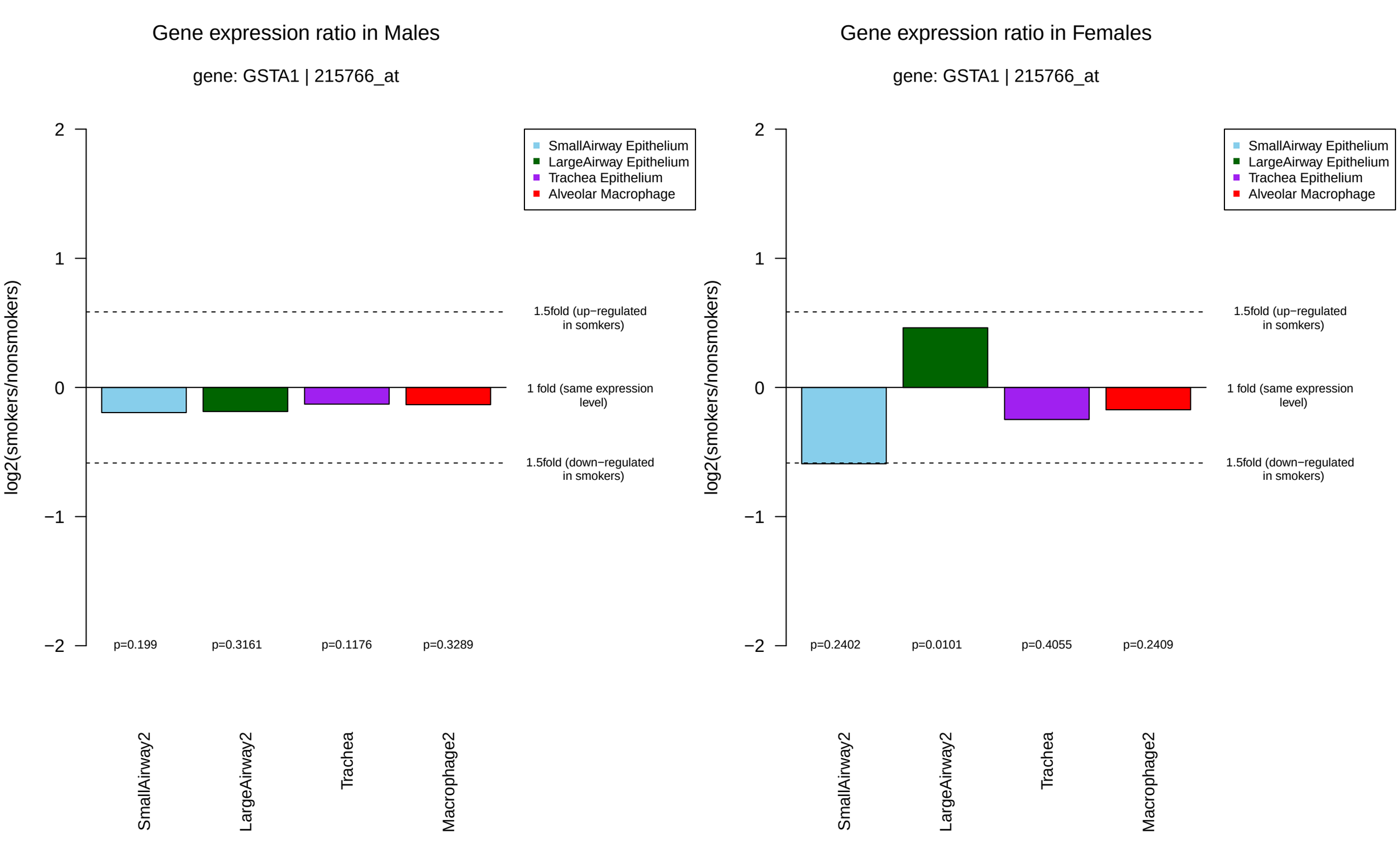
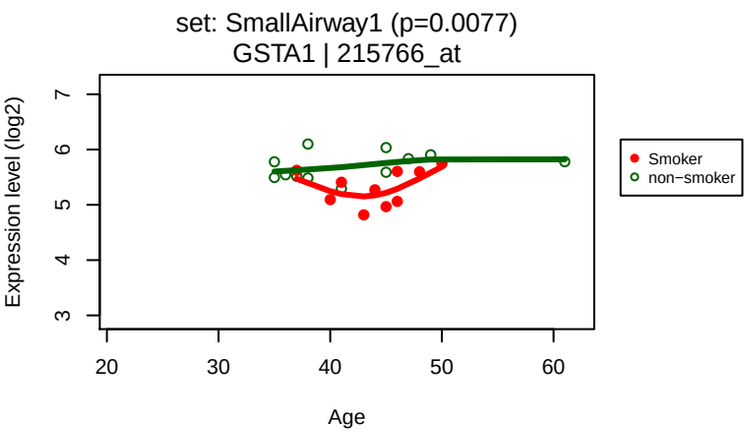
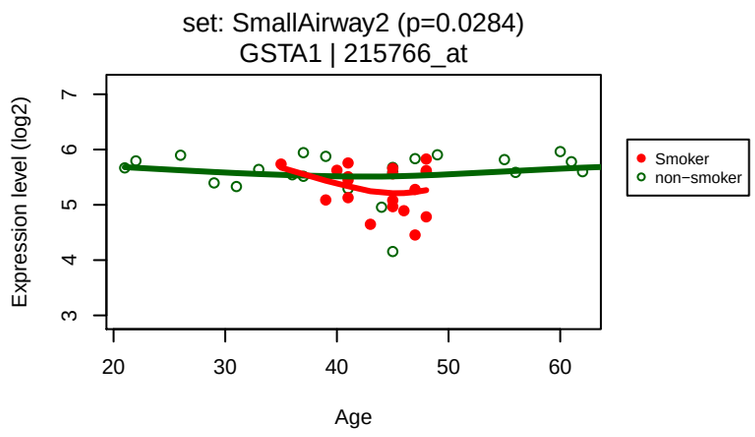
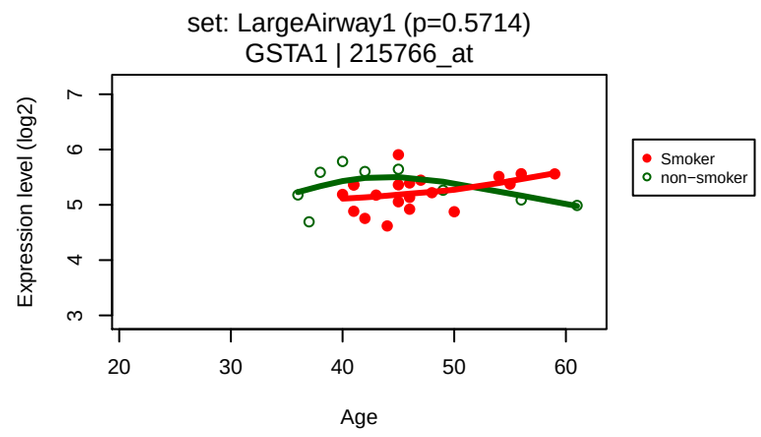

Gene expression of smokers and nonsmokers
gene: GSTA1 | 215766_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0077
p=0.0284
p=0.0023
p=0.5714
p=0.73
p=0.0653
p=0.2941
p=0.1808
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

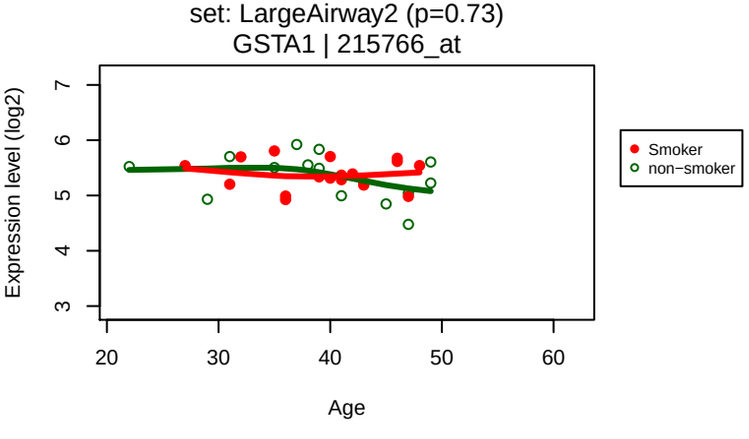

Gene expression ratio (smokers/nonsmokers)
gene: GSTA1 | 215766_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0077
p=0.0284
p=0.0023
p=0.5714
p=0.73
p=0.0653
p=0.2941
p=0.1808
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

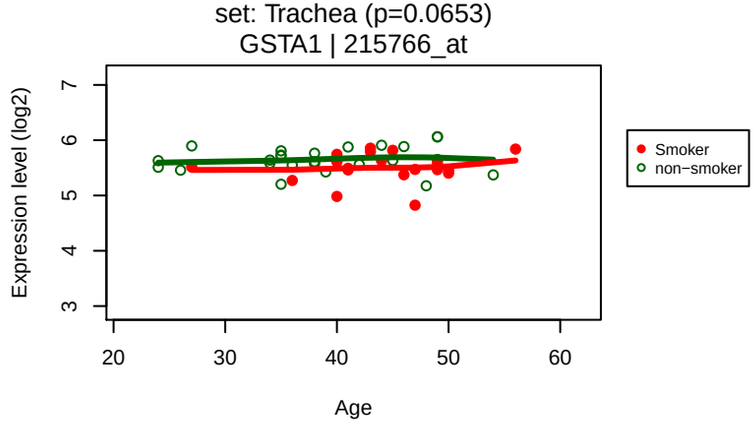

Gene expression ratio in Males
gene: GSTA1 | 215766_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.199
p=0.3161
p=0.1176
p=0.3289
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GSTA1 | 215766_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2402
p=0.0101
p=0.4055
p=0.2409
SmallAirway2
LargeAirway2
Trachea
Macrophage2

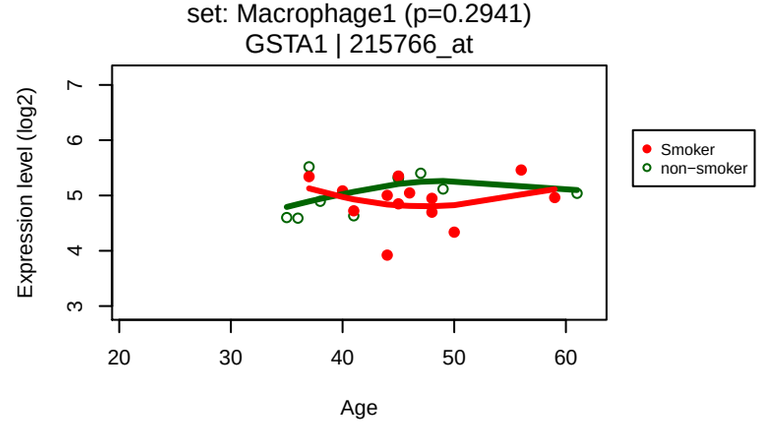

set: SmallAirway1 (p=0.0077)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker

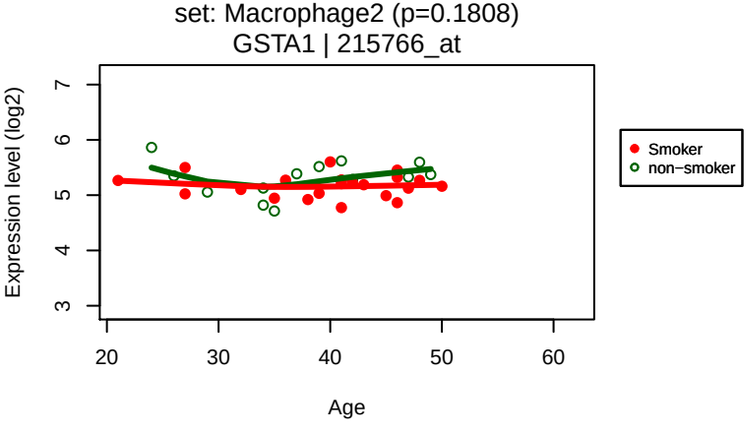

set: SmallAirway2 (p=0.0284)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker

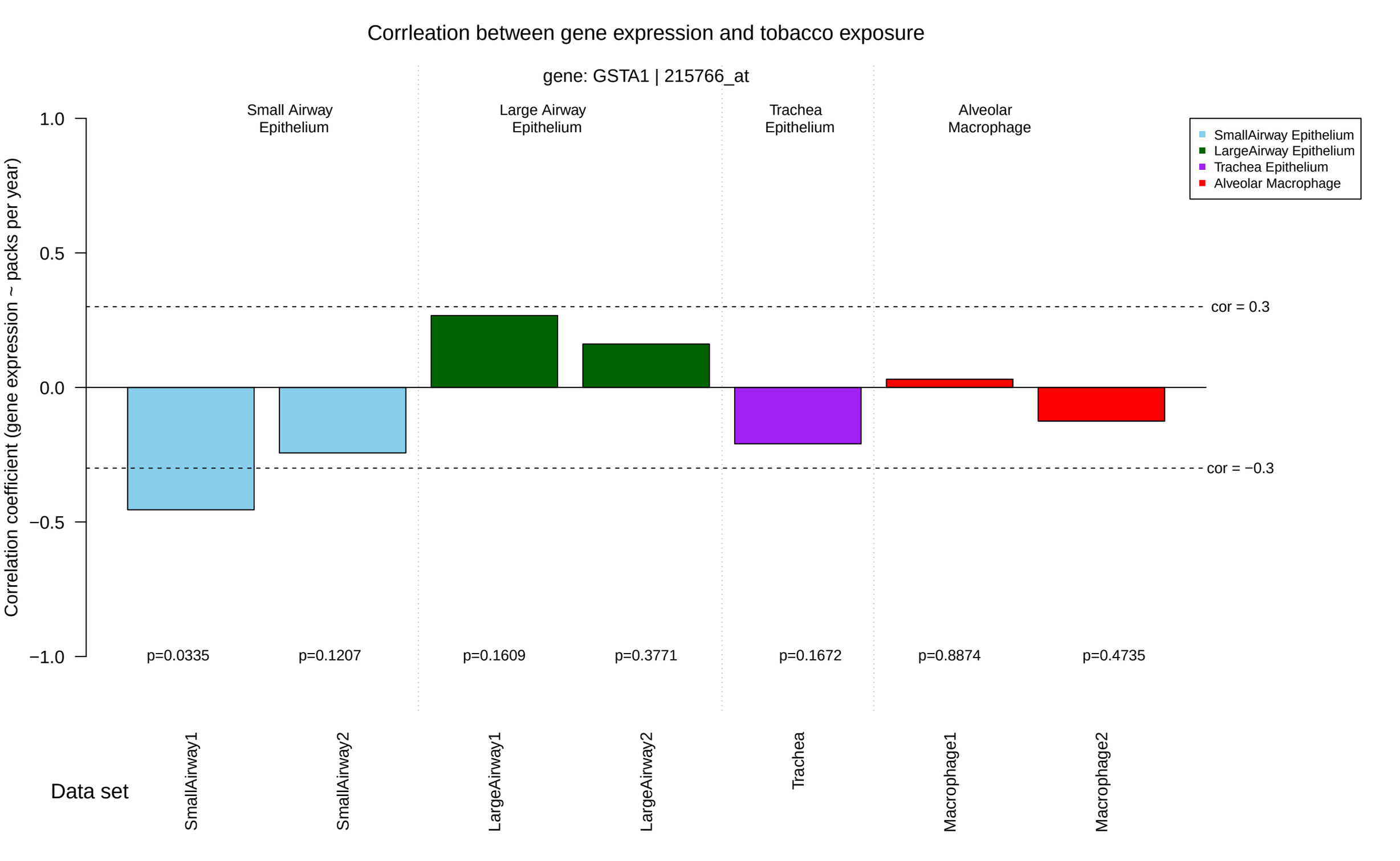

set: LargeAirway1 (p=0.5714)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.73)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0653)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2941)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1808)
GSTA1 | 215766_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GSTA1 | 215766_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0335
p=0.1207
p=0.1609
p=0.3771
p=0.1672
p=0.8874
p=0.4735
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2